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PROBLEM 4: SPORTS OR POLITICS

GitHub: https://github.com/Sagnik120/Sports_Politics_Text_Classification

Introduction

Text classification is a basic problem in Natural Language Processing (NLP), in which text data is automatically classified into predefined categories. With the ever-increasing number of online news articles and digital media, the need for automatic document classification has become more prominent than ever before.

For this project, a binary text classification model was built to classify news documents into two categories: Sport and Politics. The aim was to build a model that could classify an unseen text file into predefined categories using conventional machine learning algorithms.

Dataset Collection

The dataset employed in this project is the AG News dataset, which is a popular benchmark dataset for text classification tasks. The original dataset has four categories World, Sports, Business, Sci/Tech.

However, for this assignment, only two categories were chosen: World mapped to Politics and Sports mapped to Sport. This mapping transformed the multi-class dataset into a binary classification dataset. The dataset has about 48,000 training examples for the two chosen categories. Each example has a news headline and a brief description.

The choice of the AG News dataset was driven by the following factors: Clean and well-structured data, Balanced class distribution, Real world news articles.

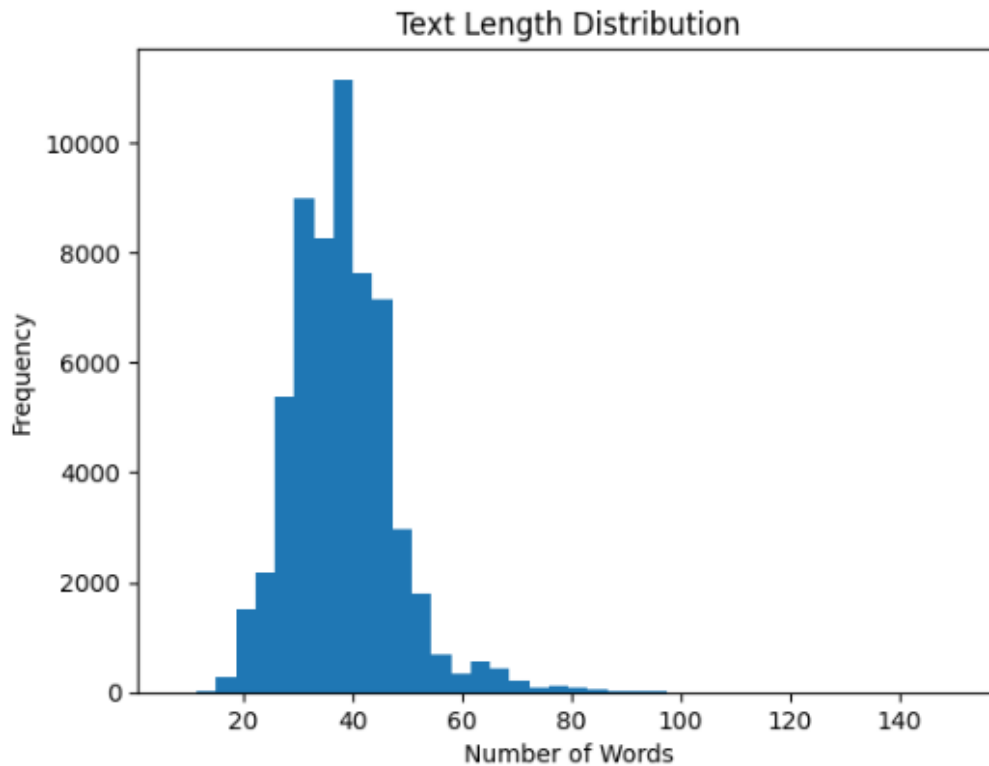
Dataset Analysis

Class Distribution

The dataset shows a balanced distribution between Sport and Politics categories. This balance ensures that the classifier does not become biased toward one class.



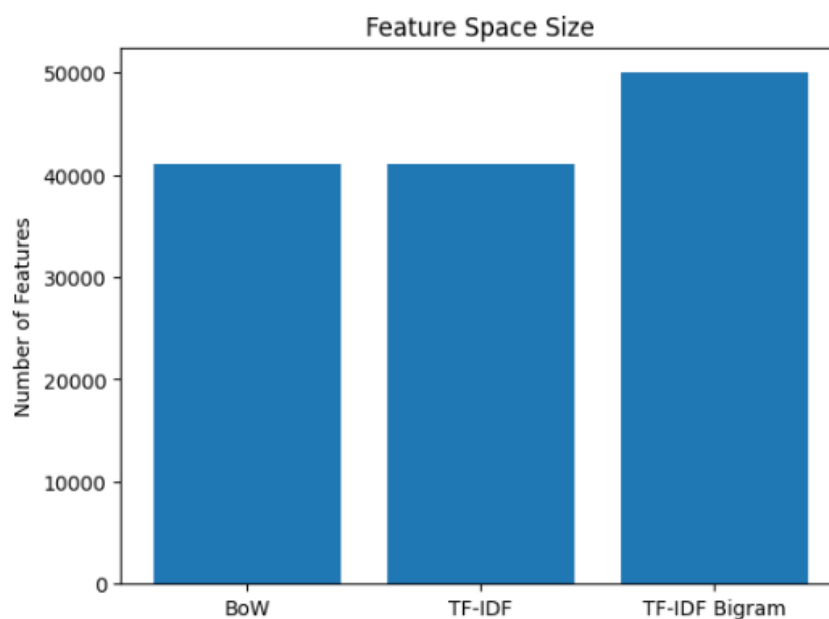
Text Length Distribution



Average length: 38.3298

From the histogram, it can be seen that the majority of the news texts have a length of 30 to 45 words, with an average of 38 words. This makes the length of the texts amenable to traditional vectorization technique

Feature Space Size Comparison



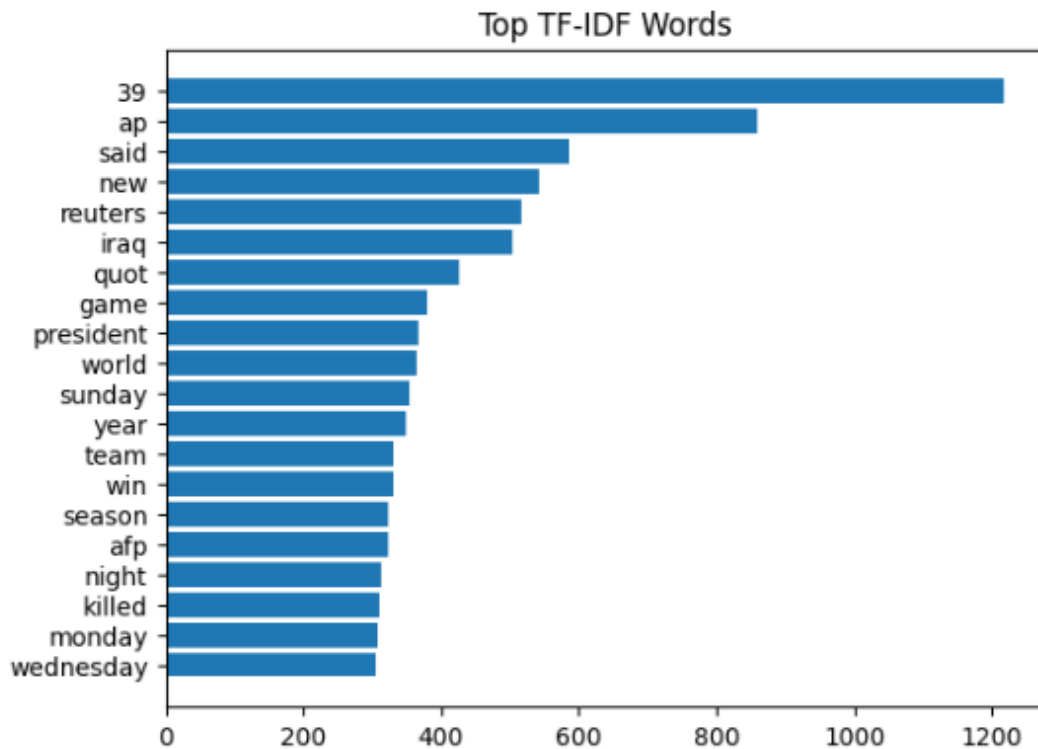


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The dimensionality of the feature space differs considerably based on the representation, which are BoW=41,000 features, TF-IDF=41,000 features, TF-IDF Bigram=50,000 features. The bigram representation causes a considerable increase in the dimensionality. Hence, the number of features was restricted to 50,000 using the max_features parameter to decrease the complexity.

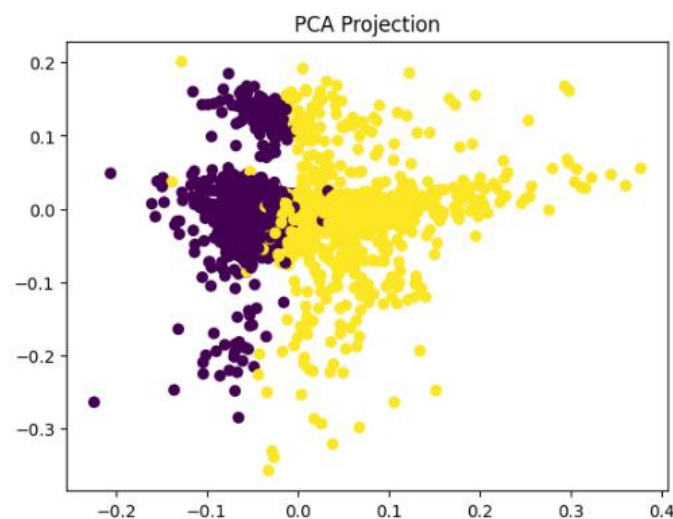
Important Words (TF-IDF)



The most influential words include domain-specific terms such as game, team, president, world

These words contribute significantly to distinguishing between Sport and Politics categories.

PCA Projection





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Principal Component Analysis was applied to project the high-dimensional TF-IDF feature space into a two-dimensional space. The projection clearly shows the clustering of the two classes. Although there is some overlap, a separation trend is apparent. This is an indication that linear classifiers can be applied to this problem.

Feature Engineering

Bag of Words (BoW)

BoW represents text as word frequency. Each document is represented as a vector of word frequencies. This approach does not consider word importance or order but is a simple and effective baseline.

TF-IDF (Unigram)

TF-IDF gives more weight to words depending on their importance. This has the effect of downplaying the importance of common words.

TF-IDF with Bigrams (n-grams)

Bigram features include pairs of words such as prime minister and world cup

Machine Learning Models

Three models were implemented and compared.

Multinomial Naive Bayes

Based on Bayes' theorem, It assumes conditional independence between words. Despite this assumption, it performs well in text classification tasks.

NB Acc: 0.97325

NB Pre: 0.9735041270344009

NB Rec: 0.97325

NB F1 : 0.9732464103742697

Linear Support Vector Machine (SVM)

SVM finds an optimal hyperplane that maximizes the margin between classes. It is particularly effective in high-dimensional sparse feature spaces.

SVM Acc: 0.9794166666666667

SVM Pre: 0.9795084259318401

SVM Rec: 0.9794166666666667

SVM F1 : 0.9794156819071713



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Random Forest

RF Acc: 0.969

RF Pre: 0.9691753673261874

RF Rec: 0.969

RF F1 : 0.9689971029515091

Quantitative Comparison

Model	Accuracy	Precision	Recall	F1
NB	0.9732	0.9735	0.9732	0.9732
SVM	0.9794	0.9795	0.9794	0.9794
RF	0.9690	0.9692	0.9690	0.9690

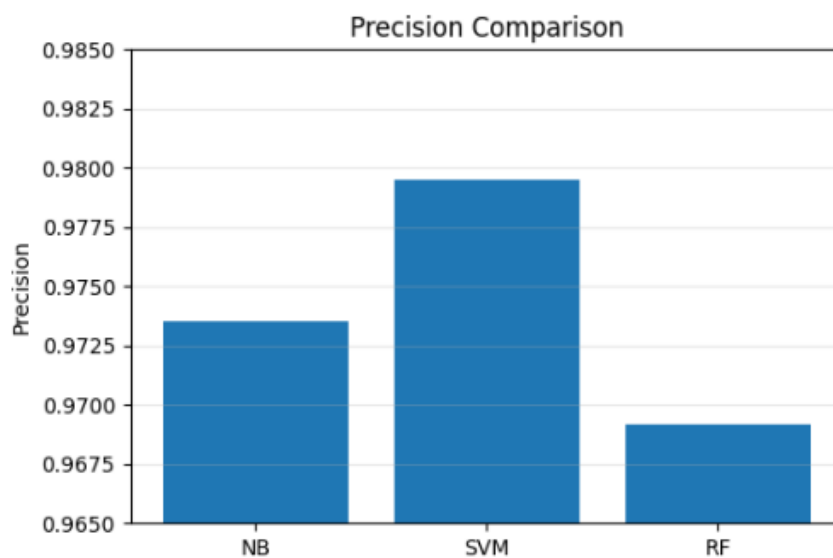
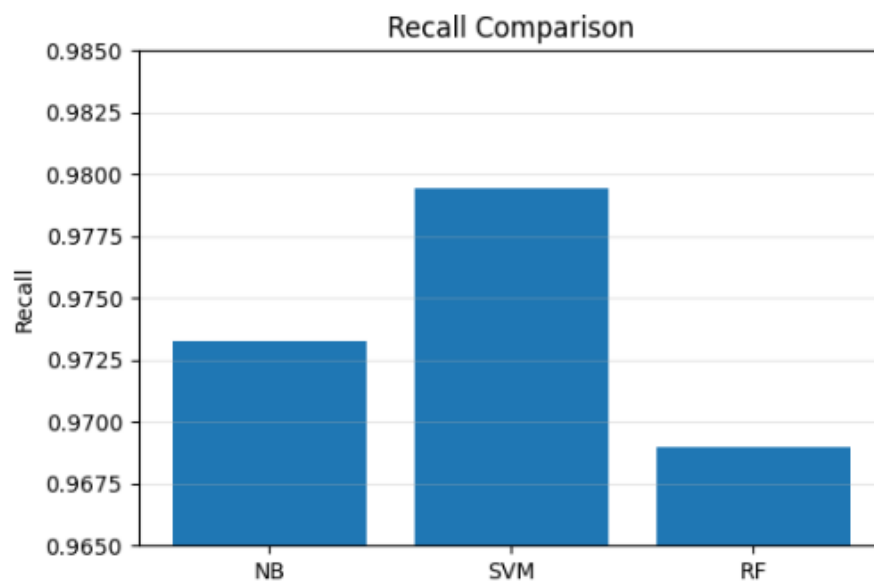
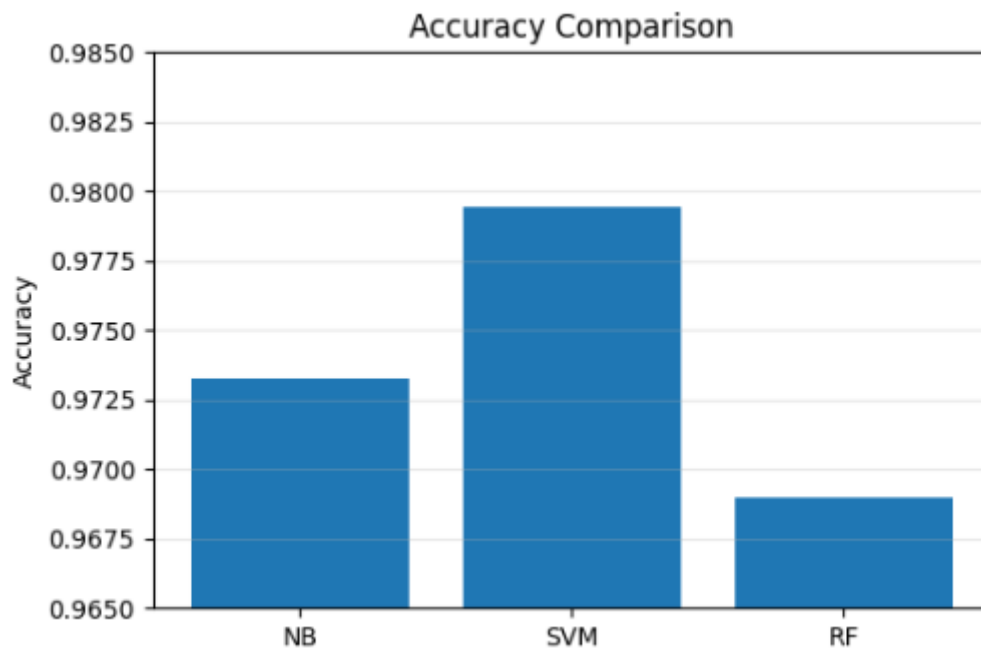
	Model	Accuracy	Precision	Recall	F1
0	NB	0.973250	0.973504	0.973250	0.973246
1	SVM	0.979417	0.979508	0.979417	0.979416
2	RF	0.969000	0.969175	0.969000	0.968997

Linear SVM performed best on all measures. Naive Bayes performed well, and Random Forest slightly worse.



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Final Classification System

A final prediction function was developed to predict the class of unseen documents. The system processes raw input text, applies transformations based on the trained TF-IDF vectorizer, and predicts the category of the input text based on the trained SVM model. This fulfills the requirement to design a classifier that can read and classify new documents.

```
# sample input
txt1 = "The government passed a new economic policy today."

out1 = pred_txt(txt1, lsv, tv2)

print("Input:", txt1)
print("Output:", out1)
```

Input: The government passed a new economic policy today.
Output: politics

Conclusion

In this project, a binary text classification system was developed to classify documents into Sport and Politics categories. Various feature engineering techniques and machine learning algorithms were explored. Among the various techniques explored, Linear SVM with TF-IDF bigram features performed the best with an accuracy of close to 98%. The results show that traditional machine learning algorithms and appropriate feature engineering can be highly effective for text classification problems.